Master's degree ICT Internet Multimedia Engineering

LAB 7: ANOMALY DETECTION

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DIPARTIMENTO DI INGEGNERIA DELL'INFORMAZIONE

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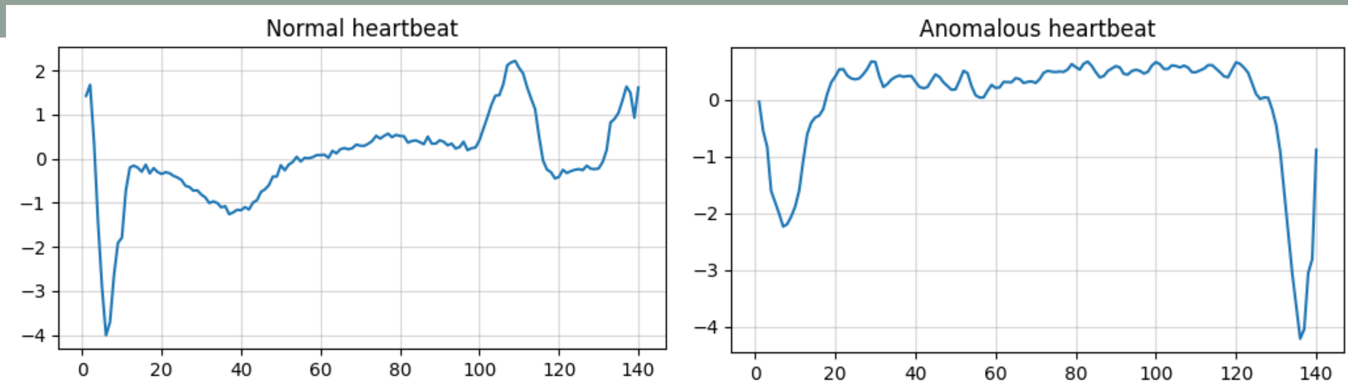
Lab 7



- The challenge:
  - Anomaly detection of heartbeats
- The ECG5000 dataset:
  - ECG5000 is derived from one record of the *BIDMC Congestive Heart Failure Database* which contains **20-hour long ECG**
  - The considered patient has **severe congestive heart failure**
  - From the record **individual heartbeats** are extracted and resampled to **140 samples**, resulting in 5,000 ECG
  - The ECG are labeled into 5 classes: one normal and four abnormal classes
  - All four abnormal heartbeat classes are considered anomalies

Anomaly detection (Part I)

In the first part of the lab, you will perform anomaly detection using autoencoders. Specifically, you will:

- Split the dataset into: `normal_train`, `normal_val1`, `normal_val2`, `normal_test`, `anomaly_val`, and `anomaly_test`
- Train an undercomplete convolutional Autoencoder <u>only on</u> `normal_train` and use `normal_val1` to prevent overfitting
- Use the error distribution on `normal_val1` to choose a set of possible thresholds
- Select the best threshold as the one **maximizing the F1 score** on `normal_val2 + anomaly_val`
- Finally, compute metrics on `normal_test + anomaly_test`

Anomaly detection (Part II)

In the second part of the lab, you will perform anomaly detection using One Class SVM (OC-SVM) on the features learned by the autoencoder. Specifically, you will:

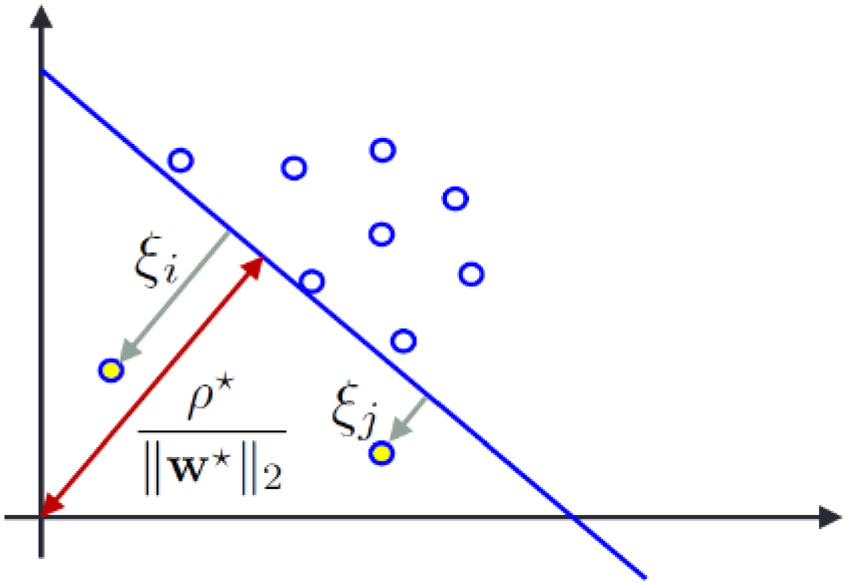
- See how to implement linear OC-SVM in scikit-learn and its shortcoming;
- Use representations obtained from the autoencoder to fit OC-SVM with different kernels;
- (Optional) Perform grid search to find the correct hyperparameters.

OC-SVM formulation

$$\min_{\mathbf{w},b} \frac{1}{2}\|\mathbf{w}\|_2^2 + \frac{1}{\boxed{\nu} N}\boxed{\sum_{i=1}^{N}\xi_i} - \rho$$

$$\text{s.t. } \mathbf{w}^T\mathbf{x}_i \geq \rho - \xi_i$$

$$\xi_i \geq 0\,,\ i = 1, \ldots, N$$



OC-SVM dual formulation

$$\max_{\alpha_1,\ldots,\alpha_N} \frac{1}{2}\sum_i\sum_j \alpha_i\alpha_j < \mathbf{x}_i, \mathbf{x}_j >$$

$$\text{subject to: } 0 \leq \alpha_i \leq \frac{1}{\nu N}, \; i = 1, \ldots, N$$

$$\sum_i \alpha_i = 1$$

$$\mathbf{w}^\star = \sum_{i:\alpha_i>0} \alpha_i \mathbf{x}_i$$

OC-SVM dual formulation

- The decision function will be:

$$f(\mathbf{x}) = \mathrm{sgn}(\mathbf{w}^\star \cdot \mathbf{x} - \rho^\star)$$

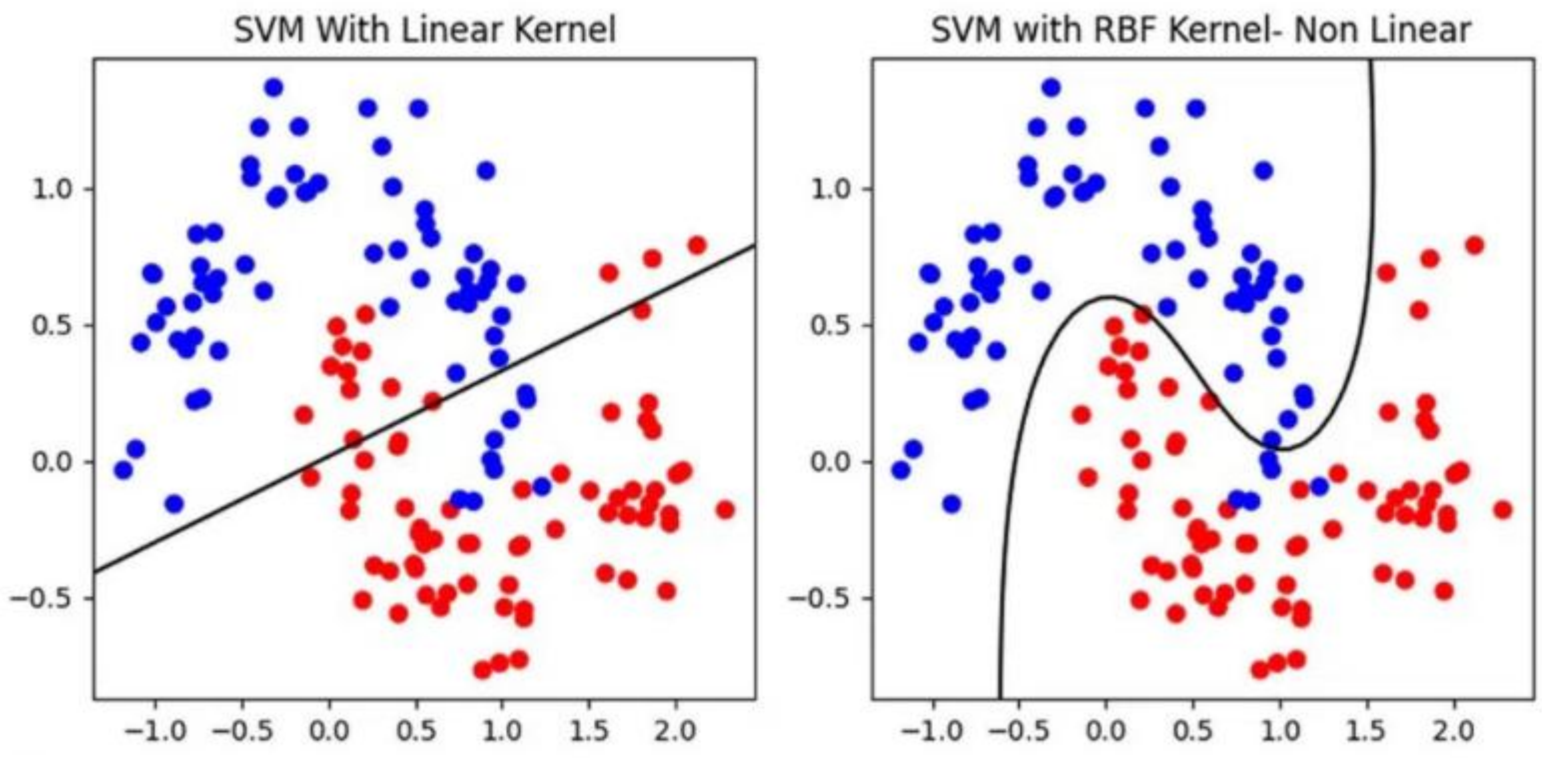
- where

$$\mathrm{sgn}(z) = \begin{cases} +1 & z \geq 0 \\ -1 & \text{otherwise} \end{cases}$$

normal behavior

outlier

Non-linear Boundaries



Non-linear boundaries and kernels

$$K(\mathbf{x}_i, \mathbf{x}_j) = \phi(\mathbf{x}_i)^T \phi(\mathbf{x}_j)$$

$$\boxed{\begin{gathered}\max_{\alpha_1,\ldots,\alpha_N} \frac{1}{2} \sum_i \sum_j \alpha_i \alpha_j K(\mathbf{x}_i, \mathbf{x}_j) \\ \text{subject to: } 0 \leq \alpha_i \leq \frac{1}{\nu N}, \; i = 1, \ldots, N \\ \sum_i \alpha_i = 1\end{gathered}}$$

Non-linear boundaries and kernels

$$\mathbf{w}^\star = \sum_{i:\alpha_i>0} \alpha_i \phi(\mathbf{x}_i)$$

The decision function becomes:

$$f(\mathbf{x}) = \mathrm{sgn}({\mathbf{w}^\star}^T \phi(\mathbf{x}) - \rho^\star) = \mathrm{sgn}\left(\sum_{i:\alpha_i>0} \alpha_i \phi(\mathbf{x}_i)^T \phi(\mathbf{x}) - \rho^\star\right)$$

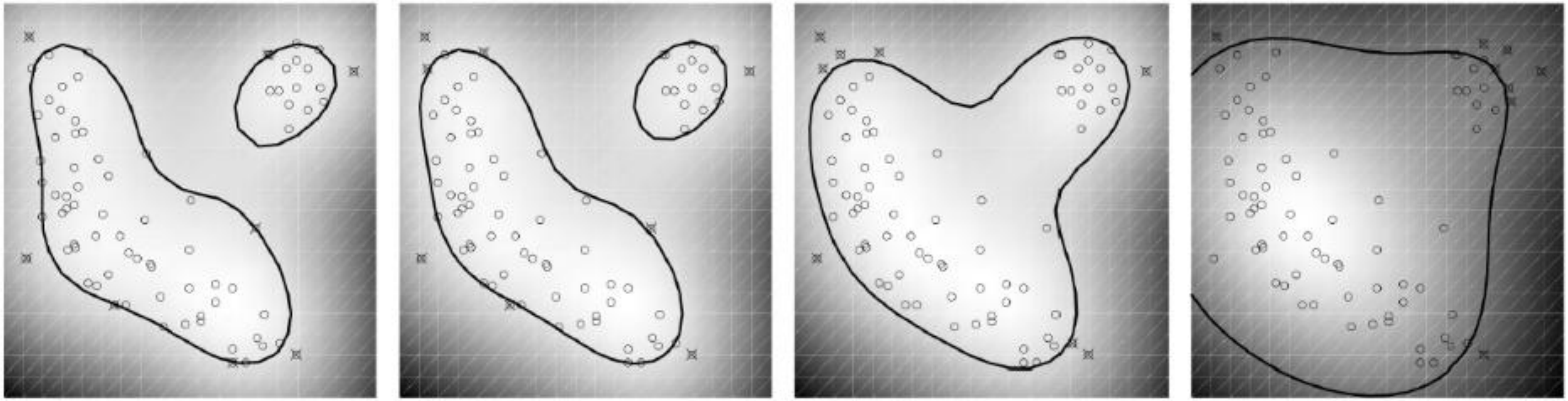
$$= \mathrm{sgn}\left(\sum_{i:\alpha_i>0} \alpha_i K(\mathbf{x}_i, \mathbf{x}) - \rho^\star\right)$$

Common Kernels

- Linear $K(\mathbf{x}, \mathbf{x}') = \mathbf{x} \cdot \mathbf{x}'$
- Polynomial $K(\mathbf{x}, \mathbf{x}') = (\mathbf{x} \cdot \mathbf{x}' + r)^d$
- Radial basis $K(\mathbf{x}, \mathbf{x}') = \exp(-\gamma \|\mathbf{x} - \mathbf{x}'\|^2)$
- Sigmoid $K(\mathbf{x}, \mathbf{x}') = \tanh(a\mathbf{x} \cdot \mathbf{x}' + b)$

RBF Kernel

- Gaussian Kernel with parameter $\gamma = \frac{1}{2\sigma^2} = 0.5$



$\nu = 0.1$ $\nu = 0.2$ $\nu = 0.4$ $\nu = 1$

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